METRIC
MIL-M-24736/1(NAVY)
31 August 1989

MILITARY SPECIFICATION SHEET

MULTIPLEXERS, DEMULTIPLEXERS, MULTIPLEXERS/DEMULTIPLEXERS (MULDEMS), MULTIPLEXER, TIME-DIVISION, FIBER OPTIC INTERFACEABLE, SHIPBOARD, 8:1 CHANNELS, 311 MBPS PER INPUT CHANNEL (METRIC)

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-M-24736.

CLASSIFICATION

Equipment function: M (Time-division multiplexer).
Signal type: A (Electrical).
Channel configuration: A (1 to 16 channels).
Analog subchannel class: J (Not applicable).
Digital subchannel class: H (Not greater than 1 gigabit per second).
Style: A (Microcircuit: integrated circuit (IC) or hybrid, package and mounting technique).

AMSC N/A
FSC 6008
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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RELIABILITY: MTBF greater than 40,000 hours.

DESIGN AND CONSTRUCTION

Case: 52-pin leadless chip carrier, quad flat pack.
4.9 cm X 4.9 cm X 1.5 cm. See figure 1 for pin diagram.

GENERAL PERFORMANCE CHARACTERISTICS

Multiplexers: 8 input channels, 1 output channel.

Analog subchannel input interface element: N/A.

Digital subchannel input interface element: maximum signal rate of 311.04 megabits per second (Mbps) per input channel.

Time-division multiplexer/control element: The output information shall be routed and organized in accordance with the frame structure shown on figure 2.

Multiplexer output interface element: The composite output data rate shall be framed with the appropriate control information at a maximum line rate of 2488.320 Mbps.

DETAILED PERFORMANCE CHARACTERISTICS

Prime power (dc): minus 5.2 volts, minus 2.0 volts.

Dc prime power variation:
plus or minus 0.3 volt for 5.2-volt supply;
plus or minus 0.1 volt for 2.0-volt supply.

Current load (dc): 0.4 to 0.7 amp for minus 5.2-volt supply;
0.1 to 0.3 amp for minus 2.0-volt supply.

Alarm functions: N/A.
Framing: N/A.
Error control: N/A.
Interchannel crosstalk: N/A.
Optical performance: N/A.
Channel performance:

Analog subchannels: N/A.
Digital channels:
Input signal voltage, high: minus 1.0 to minus 0.6 volt;
Input signal voltage, low: minus 2.0 to minus 1.6 volts.

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Interface characteristics:

Analog subchannels: N/A.

Digital channels: Signal format shall be in accordance with figure 2. The rise and fall times, and the clock and data relationship, shall be in accordance with figure 3. The data rate shall be a maximum of 311.04 Mbps per input channel, and a maximum output line rate of 2488.320 Mbps.

Multiplexed signal interface:

TDM signal output:
Electrical interface:
Output signal voltage, high: minus 0.9 to minus 0.6 volts.
Output signal voltage, low: minus 2.0 to minus 1.6 volts.
Optical interface: N/A

TDDM signal input: N/A.

Timing signals: 2.5 Gigahertz (GHz) clock. The clock input signal level shall be from 0.6 to 1.2 volts. The clock output signal level shall be 0.4 volts (peak to peak), centered at minus 1.5 volts.

Maintenance orderwire: N/A.
Other interfaces: N/A.

ENVIRONMENTAL REQUIREMENTS

Temperature range: 1 (operating - 0 to plus 85°C; storage - minus 62 to plus 85°C).
Nuclear radiation resistance: N/A.

MECHANICAL PROPERTIES:

Connector pull-out force: N/A.
Cable pull-out force: N/A.
Cable twist: N/A.
Cable seal flexing: N/A.
Fiber dynamic strength: N/A.

ELECTROMAGNETIC INTERFERENCE SUPPRESSION: N/A.

SAFETY:

Laser safety: N/A.

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SECURITY: N/A.

ENVIRONMENTAL TESTS

Explosive atmosphere: N/A.

Electromagnetic interference suppression: N/A.

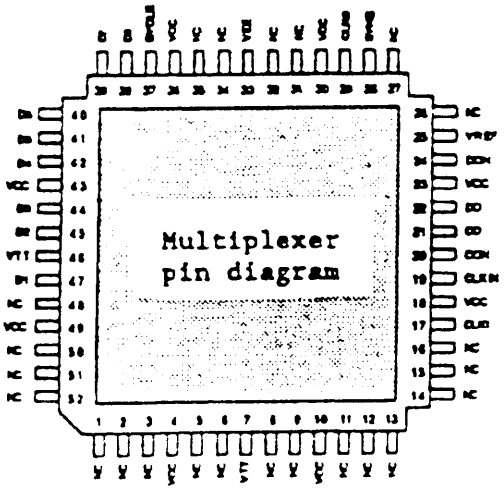
PART OR IDENTIFYING NUMBER: D24736/01-001.

PART DESIGNATOR: MAAJHA.

Review activities:
Navy - EC, YD

Preparing activity:
Navy - SH
(Project 6008-N001-01)

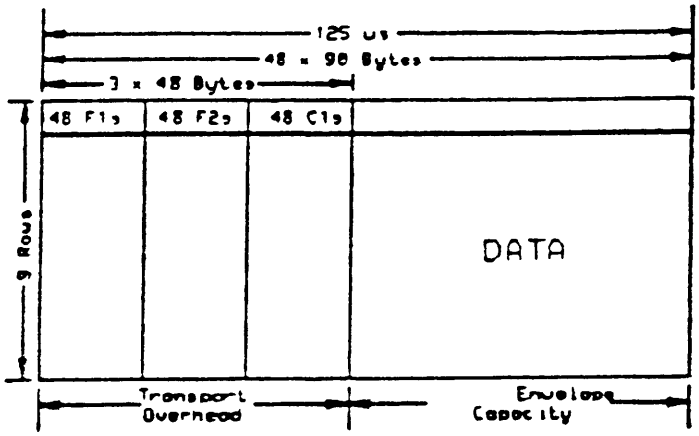
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Name	Function
D_1 - D_8	1:8 Demux parallel ECL inputs
CLKI, CLKIN	High speed differential clock inputs
SYCLK	Divide by 8 clock ECL input
DO, DON	High speed 8:1 MUX serial data output
CO, CON	High speed differential clock output
CLK8	8:1 Mux divide by 8 Clock (ECL Output)
SYNC	8:1 Mux Phase Alignment Enable (ECL Input)
NC	No Connect

NOTE:
1. Tabs indicate position of leadless contact only and are not real structures.

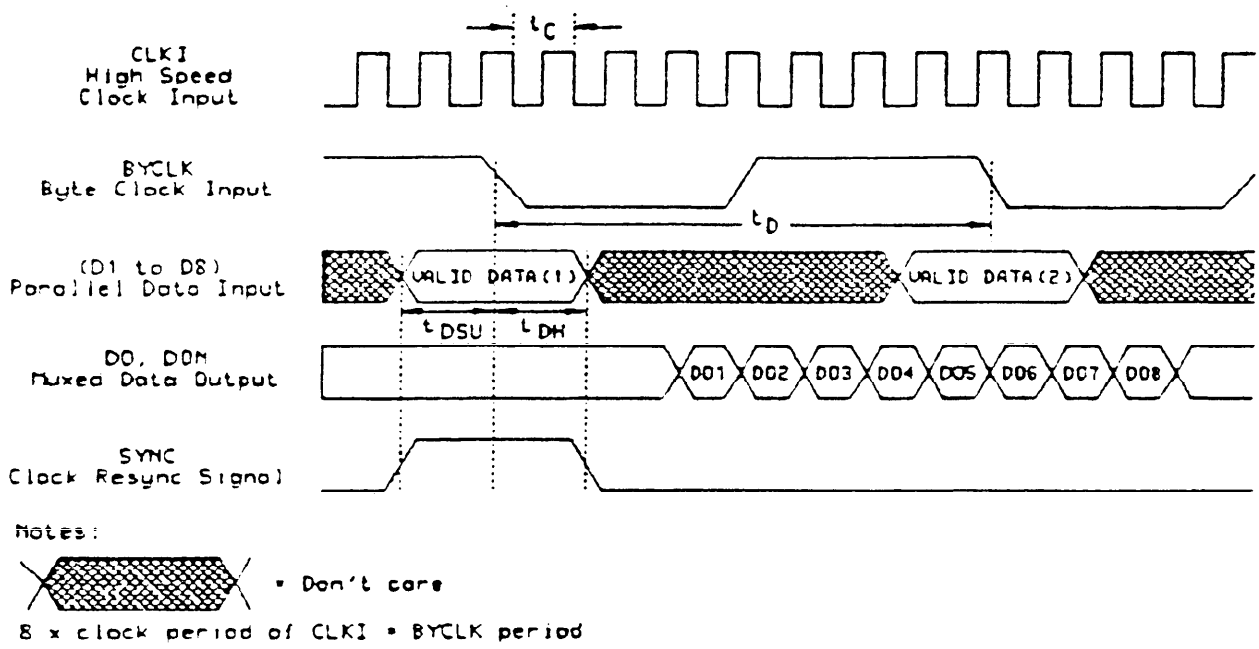
FIGURE 1. Multiplexer pin diagram.



NOTE:
1. F1 = 11110110, F2 = 00101000, C1 = Frame identification sequence.

FIGURE 2. Frame structure.

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Parameter	Description	Guaranteed Worst Case	Units
t_C	CLK period	8.4	ns
t_D	Byte clock period	3.2	ns
t_{DSU}	Parallel data set-up time	1.8	ns
t_{DH}	Data hold time	0.5	ns

FIGURE 3. Clock and data relationship.

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